

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285024

Kind Code

A1

Publication Date

September 11, 2025

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Method for Generating a Training Dataset

Abstract

A computer-implemented method for generating a training dataset to train an artificial intelligence for operating a measuring device, in particular a wall diagnostic device, includes (i) recording sensor data of at least one sensor unit of a measuring device, (ii) performing a position determination of the measuring device relative to the wall and generating position-specific sensor data using a position determination system, (iii) labeling the position-specific sensor data taking into account ground truth information and generating labeled sensor data, and (iv) aggregating the labeled sensor data into a training dataset. Also disclosed is a method for training an artificial intelligence.

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Family ID: 94820602

Appl. No.: 19/072674

Filed: March 06, 2025

Foreign Application Priority Data

DE 10 2024 202 232.2

Mar. 11, 2024

Publication Classification

Int. Cl.: G06N20/00 (20190101); G01S17/06 (20060101)

U.S. Cl.:

Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 202 232.2, filed on Mar. 11, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.

[0002] The present disclosure relates to a method for generating a training dataset for training an artificial intelligence for a wall diagnostic device. The disclosure furthermore relates to a method for training an artificial intelligence for a wall diagnostic device.

BACKGROUND

[0003] Diagnostic devices for performing wall diagnostics and detecting objects formed in the walls are known from prior art.

[0004] It is an object of the present disclosure to provide an improved method for generating a training dataset for training an artificial intelligence of a wall diagnostic device, as well as an improved method for training such an artificial intelligence.

[0005] The object is achieved by the method set forth below. Advantageous embodiments are subject-matter also set forth below.

SUMMARY

[0006] According to one aspect, a computer-implemented method is provided for generating a training dataset for training an artificial intelligence to operate a measuring device, in particular a wall diagnostic device, wherein the method comprises: [0007] recording sensor data of at least one sensor unit of a measuring device, wherein the sensor data represents a wall to be diagnosed; [0008] performing a position determination of the measuring device relative to the wall and generating position-specific sensor data using a position determination system, wherein position information is associated with each piece of sensor data recorded in the position determination, and wherein the position information defines a position of the measuring device relative to the wall, in which the measuring device was positioned relative to the wall at a time the sensor data was recorded; [0009] labeling the position-specific sensor data taking into account ground truth information and generating labeled sensor data, wherein the ground truth information comprises at least one piece of information regarding a presence of the object in the wall; and aggregating the labeled sensor data into a training dataset.

[0010] This may achieve the technical advantage of providing an improved method for generating a training dataset to train an artificial intelligence to operate a measuring device, in particular a wall diagnostic device. For this purpose, sensor data is initially recorded using at least one sensor unit of a measuring device, in particular a wall diagnostic device. The recorded sensor data here depicts a wall to be diagnosed and, if present, objects disposed in the wall.

[0011] Further, a position determination of the measuring device relative to the wall is performed and position-specific sensor data is generated by a position determination system. The position-specific sensor data includes the information from the sensor data recorded and also includes position information, wherein the position information defines a position of the measuring device relative to the wall in which the measuring device was positioned at the time the sensor data was recorded.

[0012] Subsequently, the position-specific sensor data is labeled. This takes into account ground truth information. The ground truth information comprises at least one piece of information regarding a presence of the object in the wall. The ground truth information thus describes the actual presence of an object in the wall. The labeling of the position-bearing sensor data relates here to a labeling of the sensor data with respect to the respective presence or the position of the

object.

[0013] By labeling, the sensor data is thus characterized at least in that the respectively marked piece of sensor data either depicts an object disposed in the wall or not. The position determination of the measuring device at the time the sensor data is recorded allows for labeling of the sensor data with respect to the presence of the object in the wall.

[0014] The ground truth information describes the locations in the wall at which the respective object is positioned. Mapping this information to the sensor data recorded by the measuring device requires position-specific information for the sensor data defining a position in which the measuring device was positioned at the time the sensor data was recorded.

[0015] By way of the corresponding labeled sensor data of the training dataset, artificial intelligence training of the measuring device, in particular the wall diagnostic device, is possible, in which the artificial intelligence is trained to detect an object disposed in the wall based on the sensor data of the measuring device. Detection includes detecting whether or not an object is disposed at a position on the wall.

[0016] According to one embodiment, the ground truth information comprises classification information regarding an object position and/or an object type of the object, and/or object depth information regarding an object depth within the wall, and/or object extension information regarding an object extension of the object.

[0017] By doing so, the technical advantage may be achieved that additional information may be incorporated into the training dataset by considering the additional classification information regarding the object position, the object type, the object depth information, and/or the object extension of the object. This additional information allows additional training of the artificial intelligence, which can thereby be trained to determine, based on the sensor data of the measuring device, an object position, an object type, an object depth and/or an object extension of the detected object.

[0018] In this case, the corresponding ground truth information describes the actual object position of the object in the wall, the actual object type of the object, the actual object depth of the object in the wall, and the actual object extension of the object, respectively.

[0019] The ground truth information indicated may be integrated into the training dataset or the recorded sensor data in the case of actual knowledge of the object disposed in the wall.

[0020] In the broadest sense of the application, the ground truth information describes the actual present state of the wall to be examined.

[0021] According to one embodiment, the position determination system comprises at least one camera sensor and at least one position mark, wherein the position mark is formed on a surface of the wall, wherein the camera sensor is arranged in a predefined perspective positioning towards the wall, and wherein determining the position of the measuring device relative to the wall comprises:

[0022] recording camera data of the camera sensor while recording the sensor data of the sensor unit of the measuring device, wherein the camera data depicts the measuring device positioned on the wall and the position marker disposed on the wall; [0023] determining a relative position of the measuring device relative to the position marker based on the camera data; and [0024] determining the position of the measuring device relative to the wall based on the relative position.

[0025] This may achieve the technical advantage that the position of the measuring device relative to the wall can be precisely determined. For this purpose, camera data of a camera sensor is first recorded. The camera data thereby depicts the measuring device disposed on the wall during the recording of the sensor data. Further, the camera data depicts a position marker formed on the wall. Thus, based on the camera data and taking into account the position marker, a time-resolved position of the measuring device relative to the position marker may be determined.

[0026] Finally, a time-resolved position of the measuring device relative to the wall can be determined based thereon. The procedure allows for precise positioning of the measuring device relative to the wall. Using the timestamps of the camera data, the position of the measuring device

relative to the wall may be determined precisely for a plurality of times during recording of the sensor data by the measuring device.

[0027] The method in which performing the position determination further comprises: [0028] performing a time synchronization between recording the sensor data of the sensor unit and recording the camera data of the camera sensor; and [0029] determining the position information of the measuring device relative to the wall for each piece of sensor data of the sensor unit, taking into account the time synchronization.

[0030] This may achieve the technical advantage of enabling precise generation of the position-specific sensor data. By considering the timestamps of the camera data, the respective positions of the measuring device relative to the wall can be determined for a plurality of times. The time stamps of the sensor data may be used to determine the time at which a particular piece of sensor data was recorded by the measuring device. By synchronizing the timestamps of the sensor data and the camera data over time, the position of the measuring device relative to the wall at the time the sensor data was recorded can be determined for the sensor data being recorded. Thus, precise position information may be generated for each piece of recorded sensor data that defines where the measuring device was positioned at the time each piece of sensor data was recorded.

[0031] According to one embodiment, determining the position further comprises: [0032] determining a perspective of the camera sensor relative to the wall based on the predefined perspective position in which the camera sensor is disposed relative to the wall; [0033] performing a perspective correction on the relative position of the measuring device determined based on the camera data relative to the position marker to account for the perspective position of the camera sensor relative to the wall and determining the position of the measuring device on the wall based on the perspective correction.

[0034] This may achieve the technical advantage that, by considering the perspective of the camera sensor relative to the wall, a perspective correction of the camera data may be performed. The perspective correction may be used to clean up perspective-related distortions of the position determination of the measuring device relative to the wall based on the camera data. This allows for precise positioning relative to the wall based on the camera data.

[0035] According to one embodiment, the position marker is configured as an ArUco marker or a ChArUco marker.

[0036] This may achieve the technical advantage that the position marker configured as an ArUco mark or a ChArUco mark enables a precise determination of the relative position of the measuring device relative to the position marker based on the camera data. The ArUco marker or ChArUco marker may further be applied to the wall as a correspondingly configured marker without the need to modify the wall.

[0037] The ArUco marker or ChArUco marker known from the prior art may be used as an ArUco marker or ChArUco marker.

[0038] According to one embodiment, the position marker is formed by a marker formed on the wall and comprises: wallpaper patterns, light switches, outlets, windows, furniture.

[0039] This may achieve the technical advantage that elements already formed on the wall can be used as position markers. This may eliminate the need for external wall-mounted position markers.

[0040] According to one embodiment, the camera sensor is disposed on a positioning device, and wherein the camera sensor is positioned in the predetermined perspective position relative to the wall via the positioning device.

[0041] This may achieve the technical advantage that the camera sensor may be able to be disposed precisely in a predetermined perspective position relative to the wall by the positioning of the camera sensor of the positioning device. Because the perspective position is already specified and pre-defined, the necessary perspective corrections of the camera data can be reduced to a minimum. This facilitates the position determination of the measuring device relative to the wall based on the camera data.

[0042] According to one aspect, a training dataset for training an artificial intelligence of a wall diagnostic measuring device is provided, wherein the training dataset was generated by the method for generating a training dataset for training an artificial intelligence for operating a measuring device according to one of the preceding embodiments.

[0043] According to one aspect, a computer-implemented method for training artificial intelligence of a wall diagnostic measuring device is provided, comprising: [0044] providing a training dataset by performing the method to generate a training dataset according to one of the preceding embodiments; [0045] training the artificial intelligence based on the training dataset to perform object recognition of an object formed in a wall, wherein the object recognition comprises at least an object detection and an object classification.

[0046] Provided according to one aspect, is a computing unit configured to perform the method of generating a training dataset for training an artificial intelligence for operating a measuring device according to one of the preceding embodiments and/or the method of training an artificial intelligence of a measuring device.

[0047] Provided according to one aspect is a computer program product comprising instructions which, when the program is executed by a data processing unit, prompt it to perform the method for generating a training dataset for training an artificial intelligence to operate a measuring device according to one of the preceding embodiments, and/or the method for training an artificial intelligence of a measuring device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0048] Embodiments of the disclosure are described with reference to the following figures. The figures show:

[0049] FIG. **1** a schematic representation of a measuring device according to one embodiment;

[0050] FIG. **2** a further schematic representation of the measuring device according to a further embodiment;

[0051] FIG. **3** a further schematic representation of the measuring device according to a further embodiment;

[0052] FIG. **4** a schematic representation of a measurement of the measuring device according to one embodiment,

[0053] FIG. **5** a further schematic representation of the measuring device according to a further embodiment;

[0054] FIG. **6** a further schematic representation of the system for generating a training dataset according to a further embodiment,

[0055] FIG. **7** a schematic representation of data recorded by the measuring device;

[0056] FIG. **8** a further schematic representation of the system for generating a training dataset according to a further embodiment,

[0057] FIG. **9** a flowchart of a method for generating a training dataset according to one embodiment,

[0058] FIG. **10** a flowchart of a method for training an artificial intelligence of a measuring device according to a further embodiment, and

[0059] FIG. **11** a schematic representation of a computer program product.

DETAILED DESCRIPTION

[0060] FIG. **1** shows a schematic representation of a measuring device **100** according to one embodiment.

[0061] The present disclosure relates to a measuring device, in particular a wall diagnostic device for examining walls **105** to be worked on. Wall diagnostic devices are known from the prior art that

are used to detect objects located in walls. Such devices allow a user to examine walls to be worked on to search for objects located in the walls, in order to be able to perform planned work, for example drilling in walls, such that damage to the objects located in the walls can be avoided. [0062] In the embodiment shown, the measuring device **100** comprises a housing **150** having a handle **152** for grasping of the measuring device **100** by a user, a display unit **111** for displaying diagnostic results **109** of the wall diagnostics, and controls **154** for switching the measuring device **100** between various operating modes.

[0063] According to the disclosure, the measuring device **100** comprises at least one radar sensor unit **101**. By way of the radar sensor unit **101**, radar signals may be transmitted towards the wall **105** to be examined, and radar signals reflected by the wall **105** may be received.

[0064] For example, the radar sensor unit **101** may be configured as a narrow band radar detector device in the 2.4 GHz to 2.4835 GHz frequency range or as an ultra-wide band radar detector device in the 1.8 GHz to 5.8 GHz frequency range.

[0065] The measuring device **100** further comprises a diagnostic module **107** executable on a computing unit **151** of the measuring device **100** to perform the wall diagnostics. The diagnostic module **107** is configured to perform corresponding diagnostics on the wall to be examined based on the radar data **103** of the radar sensor unit **101**. The radar data **103** of the radar sensor unit **101** thereby depicts the wall **105** to be examined and, if applicable, objects **113** located within the wall **105**.

[0066] The wall diagnostics carried out by the diagnostic module **107** comprise at least performing object recognition. The object recognition comprises an object detection and an object classification of the object **113** located in the wall **105**. The object detection comprises at least the determination of an object position **115**. The object position describes the positioning of the object located in the wall **105** with respect to a reference system defined by the measuring device **100**. The object classification of the detected object **113** comprises at least determining an object type **117** of the detected object **113**.

[0067] The diagnostic results of the wall diagnostics determined in this way, i.e. at least the determined object position **115** and/or the determined object type **117** of the object **113** located in the wall **105**, are subsequently presented in a display unit **111** of the measuring device **100** to a user of the measuring device **100**. For example, the display unit **111** may be configured as a corresponding display, and the diagnostic results **109** may be visually displayed. Additionally, the display of the diagnostic results **109** may be supported via audible and/or haptic signals. For example, the haptic signals may be realized via corresponding vibration signals.

[0068] The object **113** may be shown in the display, for example, by way of a corresponding icon. The object **113** can be shown in the corresponding object position **115** in the display. The object extension **121** may be visualized by a corresponding size of the displayed icon. The particular object type **117** of the object **113** may be visualized with a corresponding term or color highlighting of the icon, or by a specific shape of the icon representing the object **113**.

[0069] Alternatively, the wall diagnostics may additionally comprise determining a wall type **123** in the form of a wall type classification of the wall **105** to be examined. The wall type **123** describes the respective type of the wall **105** to be examined. For example, the wall type may be associated with corresponding wall type classes, which may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or wall bricks, underfloor heating, wall heating or similar wall types found in buildings.

[0070] According to one embodiment, the diagnostic module **107** is further configured to determine an object depth **119** of the object **113** within the wall **105** based on the radar data **103**. The object depth **119** is defined by a distance of the object formed in the wall **105** to a surface of the wall **105**. The distance may be defined on the object side, for example with respect to an object surface or with respect to an object center point. The distance to the surface of the wall **105** describes a shortest distance defined by a direction perpendicular to the surface of the wall **105**.

[0071] According to one embodiment, the diagnostic module **107** is further configured to determine an object extension **121** of the object **113** in at least one predefined direction based on the radar data **103**. The object extension **121** of the object **113** describes a spatial extension of the object **113** in at least one spatial direction, preferably in two spatial directions, particularly preferably in three spatial directions. The object **113** may be described here as a one-dimensional, two-dimensional, or three-dimensional object **113**.

[0072] In conventional use, the measuring device **100** is placed on the surface of the wall **105** to be examined. Radar signals are transmitted towards the wall **105**, and radar signals reflected from the wall **105** or the objects **113** located behind it are received, via the radar sensor unit **101**. This radar data **103** of the radar sensor unit **101** is used by the diagnostic module **107** to perform the wall diagnostics described above and to determine corresponding diagnostic results **109**.

[0073] For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** of the object **113** located in the wall **105**. Alternatively or additionally, the diagnostic results **109** may comprise the wall type **123** of the wall **105** and/or the object depth **119** and/or the object extension **121** of the object **113**.

[0074] The diagnostic results **109** configured in this manner may subsequently be displayed to a user of the measuring device **100** in a display unit **111** of the measuring device **100**. The display unit **111** can be configured as a corresponding display, for example. The diagnostic results **109** may be displayed in graphical or textual form in the display unit **111**.

[0075] According to one embodiment, the measuring device **100** further comprises a motion detection unit **141**. The motion detection unit **141** may be used to detect movement of the measuring device **100** relative to the wall **105**. The motion detection unit **141** may comprise, for example, at least one roller element for this purpose. When the roller element is placed on the wall surface of the wall **105**, movement of the measuring device **100** relative to the wall **105** can be detected when the measuring device **100** moves along a direction of movement **153** by rolling the roller element. Alternatively, the motion detection unit **141** may have a different configuration by which a relative movement of the measuring device **100** relative to the wall **105** can be detected.

[0076] By moving the measuring device **100** relative to the wall **105**, radar data **103** of the radar sensor unit **101** may be recorded for a plurality of different positions of the measuring device **100** relative to the wall **105**. This enables a larger spatial area of the wall **105** to be examined than the effective range of the radar sensor unit **101**. This enables objects **113** to be recorded that have a greater spatial extent than the effective range of the radar sensor unit **101**.

[0077] While the measuring device **100** moves along the direction of movement **153**, radar data **103** of the radar sensor unit **101** may be recorded continuously. The wall diagnostics may be evaluated by the diagnostic module **107** based on this radar data **103** while the measuring device **100** is moving along the direction of movement **153**. This allows for accelerated wall diagnostics, taking into account the positioning of the measuring device **100** relative to the wall **105**.

[0078] According to its embodiment, the diagnostic module **107** is configured as a correspondingly trained artificial intelligence **125**. The artificial intelligence **125** is trained to perform the above-mentioned wall diagnostics based on the radar data **103** of the radar sensor unit **101** and to determine at least the object position **115** and the object type **117** of an object **113** located in the wall **105**. The object classification or determination of the object type **117**, respectively, comprises assigning the detected object **113** to predefined object classes.

[0079] The object classes may comprise: Metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, or other elements commonly installed in building walls.

[0080] Furthermore, the artificial intelligence **125** may be trained to determine the wall type **123** of the wall **105** to be examined at least based on the radar data **103** of the radar sensor unit **101**.

Possible wall types **123** may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or

individual bricks of the brick wall, underfloor heating, wall heating or other similar wall types commonly installed in buildings.

[0081] According to one embodiment, in addition to the radar sensor unit **101**, the measuring device **100** may comprise further additional sensors by way of which additional physical variables are detectable. For example, the measuring device **100** may comprise an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor or other sensors commonly used in wall diagnostic devices.

[0082] The diagnostic module **107**, in particular the corresponding trained artificial intelligence **125**, can be configured to perform wall diagnostics based on the radar data **103** of the radar sensor unit **101** and taking into account the additional sensor information of the further sensors described above. The additional information of the additional sensors mentioned above can particularly be used for object recognition of the objects **113** located in the walls **105**. The additional sensor information may possibly provide improved detection of the objects **113** and may possibly provide improved classification of the objects **113**.

[0083] For example, the material of the objects **113**, for example as a metallic or non-metallic material, can be particularly improved and classified by using the additional sensor information.

[0084] FIG. 2 shows another schematic representation of the measuring device **100** according to a further embodiment.

[0085] In the embodiment shown, the measuring device **100** comprises a pre-processing module **127** in addition to the diagnostic module **107**. For wall diagnostics, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. Pre-processing of the receiving radar data **103** is performed via the pre-processing module **127**. For example, via pre-processing by the pre-processing module **127**, the radar data may be brought into a corresponding data structure required for wall diagnostics by the diagnostic module **107**.

[0086] As described above, during wall diagnostics, the diagnostic module **107** generates the diagnostic results **109** described above. For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** and/or object depth **119** and/or object extension **121** of an object **113** formed in the wall **105** to be examined and/or the wall type **123** of the wall **105** to be examined. The correspondingly generated diagnostic results **109** may subsequently be displayed in the display unit **111** of the measuring device **100**.

[0087] According to one embodiment, in addition to the radar data **103** of the radar sensor unit **101**, the additional sensor information of the additional sensors described above may be considered during the wall diagnostics of diagnostic module **107**. A corresponding pre-processing of the additional sensor information may be performed accordingly by the pre-processing module **127**.

[0088] In the embodiment shown, the diagnostic module **107** comprises a wall type classification module **129** and an object recognition module **131**. The pre-processing module **127** comprises a first pre-processing module **135** and a second pre-processing module **137**. The first pre-processing module **135** comprises an S matrix reduction **155**. The second pre-processing module **137** comprises a background correction **157**, an inverse Fast Fourier transformation **159**, and a focusing and migration **161**. During the pre-processing of the radar data **103** by the pre-processing module **127**, the radar data **103** is first pre-processed by the first pre-processing module **135** and the S-matrix reduction **155** contained therein.

[0089] When doing so, the first pre-processing module **135** generates input data **133** based on the radar data **103**. The input data **133** serves as input data for the wall type classification module **129**. The wall type classification module **129** performs a wall type classification of the wall **105** to be examined based on the input data **133** and generates wall type information **139**. The wall type information **139** contains the wall type **123** of the wall **105** to be examined, as determined in the wall type classification.

[0090] Subsequently, the second pre-processing module **137** performs pre-processing based on the radar data **103** and the wall type information **139**. A background correction **157** of the radar data

103 is performed during this, taking into account the wall type **123** determined in the wall type information **139**. Depending on the wall type **123** of the wall **105** to be examined, different effects can occur on the radar data **103**.

[0091] These effects, which are primarily based on the respective wall type **123** and can affect object recognition, can be corrected by the background correction **157**. After the background correction has been performed, further pre-processing can be carried out by performing the inverse Fast Fourier transformation **159** or focusing and migration **161**, respectively, and input data **133** can be created for the object recognition module **131** once again. Based on the input data **133** provided by the second pre-processing module **137**, the object recognition module **133** performs object recognition of the object **113** located in the wall **105** to be examined and determines at least the object position **115** and the object type **117** of the respective object **113**. Additionally, the object depth **119** and the object extension **121** may be determined by the object recognition module **131**.

[0092] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.

[0093] The pre-processing is optional. Depending on the algorithm used for the diagnostic module **107**, completely unprocessed radar echoes of different frequencies may be used as radar data **103** and as input data for the diagnostic module **107**. Alternatively, radar data **103** processed via multiple steps may be used. The pre-processing steps comprise, for example, the transformation of the signals from the frequency domain to the time or distance domain, background deduction, noise removal, and normalization of the signals. For radar data **103** which is available in the form of complex numbers, only the absolute value can be processed. Alternatively or additionally, the phase information may be considered.

[0094] FIG. **3** shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0095] In the embodiment shown, the diagnostic module **107** comprises a plurality of parallel processing paths **102**. Each processing path **102** includes a pre-processing module **127**, the diagnostic module **107**, comprising the wall type classification module **129** and/or the object recognition module **131**, for example, according to the embodiment in FIG. **2**, and a post-processing module **163**.

[0096] In FIG. **3**, primarily the radar data **103** is shown as input data for the wall diagnostics. In addition to the radar data shown, however, the additional information from the additional sensors may also serve as input data for the wall diagnostics. The different information from the different types of sensors in the different parallel processing paths **102** can be processed and the corresponding wall diagnostics performed separately on the different sensor information. Upon completion of the wall diagnostics, a summary module may be used to assemble a summary of the individual partial analysis results for the diagnostic results **109** of the wall diagnostics.

[0097] Alternatively or additionally, different sub-aspects of wall diagnostics may be performed by the different processing paths **102** based on the same sensor information.

[0098] For example, the individual processing paths **102** may process different radar data **103** recorded during movement of the measuring device **100** relative to the wall **105** for different positions of the measuring device **100** relative to the wall **105**. The radar data **103**, thus representing different regions of the wall **105** and recorded sequentially in time as the measuring device **100** moves relative to the wall **105**, may then be processed in the different processing paths **102** by the modules shown.

[0099] The various processing paths perform stand-alone wall diagnostics including at least determining the object position **115** and/or the object type **117** of the object **113** located in the wall **105**.

[0100] By way of the summary module **165**, the partial results of the stand-alone wall diagnostics of the different regions of the wall **105** provided in the individual processing paths **102** can be

summarized to a contiguous diagnostic result **109**. The contiguous diagnostic result describes the wall diagnostics of a contiguous spatial area that was scanned during movement of the measuring device **100** relative to the wall **105** and depicted by the corresponding recorded radar data **103**. The parallel processing of the radar data **103** or additional sensor information **104** of the additional sensor elements in the different processing paths **102** thus enables accelerated wall diagnostics. [0101] Alternatively, various wall diagnostic functions may also be performed in the different processing paths **102**. For example, in one processing path **102**, the wall type classification and the determination of the wall type **123** of the wall **105** to be examined can be performed. In another processing path **102**, object recognition of the object **113** located in the wall can be performed. The object detection can be performed with the determination of the object position **115** and the object classification can be performed with the determination of the object type **113** in one processing path **102**.

[0102] Alternatively, the object detection and object classification may then also be performed in two separate processing paths **102**. In further processing paths **102**, the object depth determination, i.e., the determination of the object depth **119** and/or the determination of the object extension **121** may each be carried out. In the summary module **165**, the various partial results of the wall diagnostics may be summarized into corresponding diagnostic results **109**.

[0103] The diagnostic module **107** can be divided into different artificial intelligence structures **125**, as already shown in the embodiment in FIG. 2. For example, the diagnostic module **107** may comprise a wall type classification module **129** and an object recognition module **131**. The object recognition module may in turn be divided into an object detection module and an object classification module. The diagnostic module **107** may further comprise an object depth determination module and object extension module, respectively configured to determine the object depth **119** and the object extension **121**.

[0104] The respective modules may each be configured as stand-alone artificial intelligence structures **125**, for example, neural networks. Alternatively, the various modules may form portions of an overall artificial neural network that are connected to an overall neural network according to structures known from prior art.

[0105] FIG. 4 shows a schematic illustration of a measurement by the measuring device **100** according to one embodiment.

[0106] For pre-processing, the radar data **103** or the additional sensor information **104** of the remaining sensors may be normalized, in particular to numerically stabilize the subsequent steps performed by the diagnostic module **107** during the wall diagnostics. For example, an amplitude and/or offset compensation may be performed for this purpose. Moreover, to reduce interference, filtering of the radar data **103** may be carried out, and to reduce the data rate, the corresponding sensor data may be sampled. Additionally, the radar data **103** or the additional sensor information **104** may be transformed into the respectively required frequency range or time range. Methods known from the prior art can be used for this purpose.

[0107] Furthermore, the recorded radar data **103** or additional sensor information **104** may be divided into temporal or spatial windows **167**. Temporal windows **167** may be generated by recording the radar data **103** or the additional sensor information or the pre-processed radar data **103** over a fixed time interval. Spatial windows **167**, on the other hand, may be generated from a mapping of the radar data **103** or additional sensor information **104** to positions of the measuring device **100** relative to the wall **105** along the direction of movement **153**.

[0108] Graph a) of FIG. 4 shows such a data matrix resulting from the steps described above. The data matrix of the window **167** shown in graph a) shows a plurality of sensor data, which may comprise, for example, radar data **103** or additional sensor information **104** of the further sensors plotted along a frequency channel axis **171** or along a space/time axis **169**, respectively.

[0109] A width of the temporal windows **167** may be selected such that different sampling rates of the sensors may be balanced and a new window **167** may be provided frequently enough so that a

display of the diagnostic results **109** of the wall diagnostics in the display unit **111** may be shown without too great of a time delay while the measurement is being performed or shortly after the measurement by the measuring device **100** ends.

[0110] A rate of 2 to 20 windows per second of the data recording of the sensor data may be advantageous for this purpose. For spatial windows, the spatial sampling rates can be selected to achieve the desired local accuracy. Advantageously, 1 mm to 1 cm sampling rates can be used. This means that corresponding sensor data is recorded every 1 mm to 1 cm of movement of the measuring device **100** along the direction of movement **153**.

[0111] A width of the spatial windows **167** may be selected such that contiguous information relating to an object **113** is contained in a window. Advantageously, a width of the respective spatial windows **167** can be 1 cm to 20 cm. This results in 4 to 100 measured values per window **167**. This allows for further efficient algorithmic processing of the correspondingly recorded radar data **103** or additional sensor information by the diagnostic module **107**.

[0112] A further temporal window **167** or spatial window **167** can be provided as soon as one or more sampling points are available.

[0113] The diagnostic module **107** may be oriented such that a matrix corresponding to the window size of the respective spatial or temporal window **167** may be included as input data, for example of each processing path **102** of the embodiment in FIG. 3 as well. According to the embodiment in FIG. 2, the corresponding input data can comprise the respective pre-processed sensor data, i.e. radar data **103** and additional sensor information **104** of the additional sensors.

[0114] As stated above, the wall diagnostics may be performed by the diagnostic module **107** based on a correspondingly trained artificial intelligence. Alternatively, different processing paths may also be calculated by way of rule-based algorithms. In addition, within a processing path **102**, a combination of artificial intelligence and rule-based algorithm is possible in the form of a parallel operation or concatenation.

[0115] The diagnostic results **109** of the wall diagnostics may be expressed as numeric values, vectors, or matrices. Furthermore, for the object detection, the probability of detection, or for the wall type classification and/or the object classification, respectively, a probability of the specified object classes and/or wall type classifications can be indicated. The same may apply to the position and/or depth determination, for which corresponding probability values can also be indicated.

[0116] In addition to the radar data **103**, if the additional sensor information of the further sensor types is processed in a processing path **102**, these may either be merged within the artificial intelligence **125** or combined by way of rule-based combinations.

[0117] During the post-processing of each processing path **102**, of the embodiment in FIG. 3, multiple algorithm results based on multiple windows **167** may be summarized by the summary module **165**. This summary can in particular be realized by way of majority formation, sum formation or also by multiplication of successive probability values.

[0118] Furthermore, by clustering multiple results, for example, it is possible to detect which objects, of multiple detected objects located close to one another, are the same object so that they are not incorrectly detected multiple times.

[0119] Likewise, it is possible to multiplicatively apply a weighting function **177** when summarizing the results from multiple windows **167**. Advantageously, the diagnostic partial results **175** corresponding to corresponding data points in the space may be weighted with respect to positioning of the diagnostic partial results **175** relative to a center point of the respective window **167**. This is illustrated by way of example in graph b), in which the individual diagnostic partial results **175** are weighted according to the weighting function **177** shown with respect to the center point of the window **167** shown.

[0120] According to one embodiment, the results of one processing path **102** after post-processing **163** may influence the extension of another processing path **102**. Weighting parameters may be adjusted that may depend on the particular result from the processing path **102** for each window.

[0121] For example, the result of an object classification in which the object type **117** of an object located in the wall **105** is defined, can be used to increase the weight of a wall type classification, in which the wall type **123** of the respective wall **105** is determined, during the post-processing at locations without objects **113**, because the respective radar data **103** at these locations is less influenced by reflections of the objects **113**.

[0122] FIG. 5 shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0123] Graphs a) and b) of FIG. 5 show two different alternatives of joint data processing of radar data **103** and additional sensor information **104** by the diagnostic module **107**.

[0124] Graph b) illustrates a joint processing of the radar data **103** and the additional sensor information **104** of the additional sensors by the diagnostic module **107**. For this purpose, the radar data **103** and the additional sensor information **104** are collectively used as input data of the diagnostic module **107** configured as an artificial intelligence, in particular as an artificial neural network. The diagnostic module **107** here comprises multiple convolutions **108** and multiple dense layers **106**. The radar data **103** and the additional sensor information **104** are processed jointly as input data via the convolutions **108** and dense layers **106**. The aforementioned diagnostic results **109** are created as output data of the diagnostic module **107** based on this.

[0125] On the other hand, in graph b) the radar data **103** and the additional sensor information **104** are used as stand-alone input data of the diagnostic module **107**. The diagnostic module **107** becomes multiple processing paths **102**. The processing paths **102** each comprise multiple convolutions **108** and at least one dense layer **106**. In the various processing paths **102**, wall diagnostics are performed separately by the diagnostic module **107** based on the radar data **103** and the additional sensor information **104**, respectively.

[0126] In an additional concatenation layer **148**, the partial results of the partial diagnostics of the different processing paths **102** are combined and fed to a final dense layer **106**. The output data of the diagnostic module **107** corresponds to the diagnostic results **109** described above.

[0127] The correspondingly configured diagnostic module **107** is configured to perform wall diagnostics as described above, including the features described above, based on the radar data **103** and the additional sensor information **104**.

[0128] In the embodiment shown, the diagnostic module **107** is configured as an artificial neural network, in particular as a convolutional network. Corresponding network architectures with convolutions **108**, dense layers **106**, and concatenation layers **148** are known from the prior art.

[0129] FIG. 6 shows a schematic representation of a system **600** for generating a training dataset according to one embodiment.

[0130] According to the present disclosure, to generate a training dataset **143**, sensor data **103**, **104** of at least one sensor unit **101** of the measuring device **100** is first recorded. For this purpose, several measurements can be taken by a measuring device **100** or by a plurality of measuring devices **100** of a wall **105** or a plurality of different walls **105**. During the measurements **105**, the measuring device **100** may be moved along the direction of movement **153** along the walls **105** to be examined and corresponding sensor data **103**, **104** may be recorded, comprising the walls **105** to be examined and, if applicable, objects **113** disposed therein, as described above. Sensor data **103**, **104** may include radar data **103** and/or additional sensor information **104**.

[0131] During the recording of this sensor data **103**, **104** by the measuring devices **100**, position information **172** is determined for each measuring device **100** by a position determination system **145**. The position information **172** describes the positions of the measuring device **100** relative to the wall **105** in which the measuring device **100** was positioned during the recording of the sensor data **103**, **104**.

[0132] The position information **172** may thereafter be integrated into the sensor data **103**, **104** to generate position-specific sensor data thereon.

[0133] The sensor data **103**, **104** may be labeled based on the position information **172**, which

defines the position of the measuring device **100** relative to the wall **105** at the time the respective sensor data **103, 104** was recorded for each piece of sensor data **103, 104**. When labeling the sensor data **103, 104**, each piece of sensor data **103, 104** is provided with ground truth information. The ground truth information relates to at least one piece of information indicating whether the respective sensor data **103, 104** represents the object **113** disposed in the wall **105**.

[0134] This ground truth information depends, on the one hand, on whether an object **113** is disposed in the examined wall depicted by the respective piece of sensor data. Further, the ground truth information depends on whether the object **113** disposed in the wall **105** is disposed in an object position **115** that at least partially matches the position of the measuring device **100** relative to the wall **105** at the time the respective piece of sensor data **103, 104** is recorded.

[0135] Additionally, the ground truth information may further include information regarding an actual object location **115** of the object **113** and/or regarding an object type **117** of the object **113** and/or regarding an object depth **119** of the object **113** and/or regarding an object extension **121** of the object **113**. Further, the ground truth information may include information regarding a wall type **123** of the examined wall **105**.

[0136] Through labeling of the sensor data **103, 104** with respect to ground truth information, each piece of data is labeled with respect to the ground truth information, at least characterizing whether the respective piece of sensor data **103, 104** represents an object **113** disposed in the wall **105**.

[0137] The labeling of the sensor data **103, 104** corresponds to a labeling of the sensor data **103, 104** with respect to the corresponding ground truth information.

[0138] The position information **172** of the position determination system **145** may clearly define for each piece of sensor data **103, 104** the location at which the measuring device **100** was positioned relative to the wall **105** at the time the respective piece of sensor data **103, 104** was recorded. If the actual object positions **115** of the objects **113** disposed in the wall **105** are known, this enables clear and unambiguous labeling of the sensor data **103, 104** with respect to the object position **115**, i.e., whether an object **113** actually is depicted by the respective piece of sensor data **103, 104**.

[0139] The sensor data **103, 104** labeled in this way is subsequently summarized in a corresponding training dataset **143**.

[0140] Labeling the sensor data **103, 104** as well as considering the position information **172** and generating the training dataset **143** may be performed by an external computing unit **170**.

[0141] For example, the position determination system **145** may comprise a position sensor disposed on the measuring device **100**. A positioning of the measuring device **100** relative to the wall can thus be determined via the position sensor.

[0142] Preferably, the position determination system **145** comprises a camera sensor positioned externally to the measuring device **100**. Camera data may be recorded via the camera sensor, which depicts the measuring device **100** while recording the sensor data **103, 104**. A positioning of the measuring device **100** relative to the wall **105** while recording the sensor data **103, 104** can thus be determined via the camera data.

[0143] The position of the measuring device **100** relative to the wall **105** can be determined in a time-resolved manner via the timestamps of the camera data. The respective positions of the measuring device **100** relative to the wall **105** can thus be determined for a plurality of times. The time stamps of the sensor data **103, 104** may be used to determine the times at which the respective sensor data **103, 104** was recorded by the measuring device **100**.

[0144] To determine the position information **172**, a time synchronization of the camera data of the camera sensor and the sensor data **103, 104** of the measuring device **100** can further be performed. By synchronizing the respective timestamps of the sensor data **103, 104** and the camera data in time, it can be achieved that the correct position information **172** can be associated with each piece of sensor data **103, 104**.

[0145] FIG. 7 shows a further schematic representation of the system **600** for generating a training

dataset according to a further embodiment.

[0146] By way of example, graphics a), b) show two pieces of camera data, i.e. two images or two frames of the camera sensor **147** of the position determination system **145**. The images shown in graphics a), b) depict a wall **105**. A position marker **148** in the form of an ArUco/ChArUco marker **158** is disposed on the wall **105**. Further, a light switch **162** is positioned on the wall, which is also defined as a position marker **148**. A door **160** is disposed adjacent the wall **105**, which is also defined as a position marker **148**.

[0147] A measuring device **100** is also disposed in graphic a) in the sense of the present disclosure. The measuring device **100** may be moved along the direction of movement **153** to record corresponding sensor data **103**, **104**, so as to perform wall diagnostics for the area of the wall **105** over which the measuring device **100** travels as it moves through this spatial area.

[0148] According to one embodiment, the position information **172**, i.e., the position determination of the measuring device **100** relative to the wall **105** while recording the sensor data can be carried out by using the location markers **148** shown in graphic a). Here, detection of the position marker **148** and the measuring device **100** in the camera data may be used to determine a position of the measuring device **100** relative to the position marker **148**. The relative position determined between the measuring device **100** and the position marker **148** in this way allows the position of the measuring device **100** relative to the wall **105** to be determined. The position information **172** may be created based on this, respectively.

[0149] Graphic b) furthermore graphically represents a ground truth information regarding an object **113** disposed in the wall in the area of the position marker **148** positioned in the area of graphic a) in the form of the ArUco/ChArUco marker **158**. The ground truth information includes at least the positioning of the object **113** in the wall **105**.

[0150] In graphic b), a cable is indicated as an object **113**. This is to indicate that the corresponding information, i.e., the object position **115**, the object type **117**, and other information regarding the object **113** is known as ground truth information.

[0151] Via the ground truth information shown, a corresponding labeling of the sensor data **103**, **104** recorded by the measuring device **100** in the area of the ArUco/ChArUco marker **158** shown in the area of graphic a) can be used to generate a corresponding training dataset.

[0152] FIG. 8 shows a schematic representation of data recorded by the measuring device **100**.

[0153] Graphics a), b), c) show different measurement surfaces **164** defining spatial areas where sensor data **103**, **104** was recorded by moving the measuring device **100** relative to the wall **105**. For example, the measurement surfaces **164** may describe the surface of the ArUco/ChArUco marker **158**.

[0154] Within the measurement surface **164**, the measuring device **100** is moved in multiple measurement paths **166** to record the sensor data **103**, **104**. Along the measurement paths **166**, sensor data **103**, **104** is recorded at different measurement points **168**.

[0155] The measurement paths **166** may be arranged largely parallel to one another, as shown in graphic a). The measurement paths may also be arranged crossing one another, as shown in graphic b).

[0156] The position information **127** may be integrated with the sensor data **103**, **104** recorded along the measurement path **166** at the measurement points **168** such that the respective sensor data **103**, **104** at each measurement point **168** of the different measurement paths **166** is used to determine the position of the measuring device **100** to record the respective piece of measurement data **103**, **104**. If the measurement paths **166** intersect, the position information **162** already determined may be reused. Moreover, if the measurement points **168** are close together, the corresponding position information may be determined based on the position formations of the closely disposed measurement points **168**.

[0157] Such fine mesh scanning of the measurement surface **164** in the measurement paths **166** shown may additionally generate a background correction based on the sensor data **103**, **104**

because the measurement points **168** are spaced close together with respect to one other. The background correction takes into account the measurement signals that are only reflected by the wall **105** and show no influence of an object disposed in the wall **105**. An object determination can be carried out based on these in which the change in the recorded sensor data **103**, **104** of the different measurement points **168** is determined.

[0158] FIG. **9** shows a further schematic representation of the system **700** for generating a training dataset **143** according to a further embodiment.

[0159] In the embodiment shown, the position determination system **145** comprises a camera sensor **147**. The camera sensor **147** is disposed on a positioning device **156**. The positioning device **156**, and in particular the camera sensor **147**, are disposed a distance from the wall **105**. A position marker **149** in the form of an ArUco/Ch ArUco marker **158** is disposed on the wall **105**. Further, a measuring device **100** is shown in FIG. **9** on the wall **105** in the area of the ArUco/ChArUco marker **158**.

[0160] FIG. **9** illustrates that, due to the distance of the camera sensor **147** from the wall **105**, a perspective correction of the camera data recorded by the camera sensor **147** for determining the position of the measuring device **100** relative to the wall **105** must be performed.

[0161] A perspective correction may be performed by the mathematical relationships shown below.

$$[00001] \begin{bmatrix} u \\ ? \\ 1 \end{bmatrix} = EKT_w^c \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad \text{? indicates text missing or illegible when filed}$$

[0162] Here, E is the unit matrix, K is the camera matrix, wherein the camera matrix comprises the intrinsic camera data of the camera sensor **147**, and T is the transformation vector describing the transformation of a world coordinate system to a camera coordinate system of the camera sensor **147**. The entries u and v represent pixel elements in the image plane. The components X_w , Y_w , Z_w represent the world coordinates of a world coordinate system. The components X_c , Y_c , Z_c represent the coordinates of the camera coordinate system of the camera sensor **147** shown in FIG. **9**.

[0163] Taking the matrix notation into account results in the following equation:

$$[00002] \begin{bmatrix} u \\ ? \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} ? & 0 & ? \\ 0 & ? & ? \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & ? \\ r_{21} & r_{22} & ? & ? \\ r_{31} & ? & ? & ? \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad \text{? indicates text missing or illegible when filed}$$

[0164] Here, f represents a focal length of the camera sensor **147**, r represents elements of a rotation matrix R, and t represents elements of a translation vector. Subsequently, a transformation calculation can be carried out to obtain an actual position of the camera sensor **147** in the world coordinate system. The world coordinate system is represented by the ArUco/ChArUco marker **158**. To convert vectors from the camera to the world system, the following transformation is required:

$$[00003] T_c^w = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^{-1} & -R^{-1}t \\ 0 & 1 \end{bmatrix}$$

[0165] Here, R is the rotation matrix. This correlation allows the real world position and location information of the camera sensor **147** to be extracted from the camera data in reference to an Aruco/ChArUco marker **158**. The entire video or only individual frames of the camera data may be used for this purpose. Further, by a further vector-based transformation, it may be possible to determine the position and location of the [0166] measuring device **100** depicted by the camera data. This results in an image-based vector relationship that is to be resolved by projection in 3D. To this end, positions or points from the camera data must be converted into positions with respect

to the world coordinate system, from which the required vector is subsequently calculated. This is done through

$$\begin{bmatrix} X_W \\ Y_W \\ Z_W \end{bmatrix} = R^{-1} (K^{-1} \begin{bmatrix} u \\ ? \\ 1 \end{bmatrix} - \begin{matrix} .\text{fwdarw.} \\ t \end{matrix}) \quad ? \text{ indicates text missing or illegible when filed}$$

[0167] Finally, one obtains vector x , which describes the transformation of projected positions of the measuring device **100** in the camera data to actual positions of the measuring device **100** in space, i.e. relative to the wall **105**. This correlation is shown graphically in FIG. **9**.

$$\begin{matrix} X_{D_w} & X_{C_w} \\ \text{[00005]} \cdot \text{fwdarw.} & \\ X_{DC_w} & = \begin{bmatrix} Y_{D_w} \\ Z_{D_w} \end{bmatrix} - \begin{bmatrix} Y_{C_w} \\ Z_{C_w} \end{bmatrix} \end{matrix}$$

[0168] Here, C_w are camera coordinates of the camera sensor **147** in the world coordinate system, D_w are device coordinates of the measuring device **100** in the world coordinate system, wherein the relation shown represents a conversion of the camera coordinates into the device coordinates of the measuring device **100**. An accuracy of determining the position of the measuring device **100** relative to the wall **105** may in this procedure depend in a sensitive manner on the quality of a calibration of the camera sensor **147** and an unambiguous detection of the position marker. It may therefore be advantageous to calibrate the camera sensor **147** prior to performing the method and to take sufficient test images to ensure that the detection of the position marker is reliable.

[0169] By performing the method described above, the position of the measuring device **100** relative to the wall **105** may be determined at each point in time. This may be utilized to associate the sensor data **103**, **104** recorded during measurements by the measuring device with these positions of the measuring device **100** relative to the wall **105**. In particular, multiple measurements may be compiled into a measurement sequence and these may be advantageously be processed together.

[0170] FIG. **10** shows a flowchart of a method **300** for generating a training dataset **143** according to one embodiment.

[0171] To generate a training dataset **143** for training an artificial intelligence **125** for operating a measuring device **100**, in particular a wall diagnostic device, sensor data **103**, **104** is initially recorded by at least one sensor unit **101** of the measuring device **100** in a first method step **301**. The sensor data **103**, **104** here depicts a wall **105** to be diagnosed.

[0172] In a further method step **303**, a position determination of the measuring device **100** relative to the wall **105** is performed and position-specific sensor data **103**, **104** is generated by a position determination system **145**. In the position determination, each recorded piece of sensor data **103**, **104** is associated with a piece of position information **172**. The position information **172** defines a position of the measuring device **100** relative to the wall **105** at which the measuring device **100** was positioned relative to the wall **105** at a time of the respective piece of sensor data **103**, **104**.

[0173] According to one embodiment, the position determination system **145** comprises at least one camera sensor **147** and at least one position marker **149**. The position marker **149** is formed on a surface of the wall **105** and the camera sensor **147** is disposed in a predefined perspective position with respect to the wall **105**.

[0174] Thus, in a method step **309**, camera data of the camera sensor **147** is first recorded during the time of recording of the sensor data **103**, **104** by the measuring device **100**, wherein the camera data depicts the measuring device **100** positioned on the wall **105** and the position marker **149** disposed on the wall **105**.

[0175] In a method step **311**, a relative position of the measuring device relative to the position marker **149** is determined based on the camera data.

[0176] In a method step **315**, a time synchronization is subsequently performed between the

recorded sensor data **103**, **104** of the measuring device **100** and the recorded camera data of the camera sensor **147**.

[0177] Following this, in a method step **317**, a position information **172** in which the position of the measuring device **100** relative to the wall **105** at the time of recording a piece of sensor data **103**, **104** is defined is determined for each piece of sensor data taking time synchronization into account.

[0178] Next, in a method step **313**, a position of the measuring device **100** relative to the wall **105** is determined.

[0179] To do this, a perspective of the camera sensor **107** relative to the wall **105** based on the predefined perspective position of the camera sensor **147** relative to the wall **105** is determined in a method step **319**.

[0180] Subsequently, in a method step **321**, a perspective correction is performed based on the relative position of the measuring device **100**, determined based on the camera data, relative to the position marker **149** to account for the perspective position of the camera sensor **147** relative to the wall **105**.

[0181] In a further method step **305**, the position-specific sensor data **103** is subsequently labeled taking ground truth information into account and labeled sensor data **103**, **104** is generated. The ground truth information comprises at least information regarding the presence of at least one object **113** in the wall **105**.

[0182] In a further method step **307**, the labeled sensor data **103**, **104** is aggregated to a training dataset **143**.

[0183] FIG. **11** shows a further flowchart of the method **300** for generating a training dataset **143** according to a further embodiment.

[0184] To train an artificial intelligence **125** of a wall diagnostic measuring device **100**, a training dataset **143** is first provided in a method step **401** by performing the method **300** to generate a training dataset **143** according to the embodiments described above.

[0185] In a further method step **403**, the artificial intelligence **125** is trained based on the training dataset **143** to perform an object recognition of an object **113** formed in a wall **105**, wherein the object recognition comprises at least an object detection.

[0186] FIG. **12** shows a schematic representation of a computer program product **500** comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method **300** for generating a training dataset **143**.

[0187] In the embodiment shown, the computer program product **500** is stored on a storage medium **501**. The storage medium **501** can in this case be any desired storage medium known from the prior art.

Claims

1. A computer-implemented method for generating a training dataset to train an artificial intelligence for operating a measuring device, comprising: recording sensor data of at least one sensor unit of a measuring device, wherein the sensor data depicts a wall to be diagnosed; performing a position determination of the measuring device relative to the wall and generating position-specific sensor data using a position determination system, wherein in the position determination, a position information is associated with each piece of recorded sensor data, and wherein the position information defines a position of the measuring device relative to the wall, in which the measuring device was positioned relative to the wall at a time the sensor data was recorded; labeling the position-specific sensor data taking into account ground truth information and generating labeled sensor data, wherein the ground truth information comprises at least information regarding a presence of at least one object in the wall; and aggregating the labeled sensor data into a training dataset.

2. The method of claim 1, wherein the ground truth information comprises classification

information regarding an object location of the object and/or an object type of the object and/or object depth information regarding an object depth within the wall and/or object extension information regarding an object extension of the object.

3. The method of claim 1, wherein the position determination system comprises at least one camera sensor and at least one position marker, wherein the position marker is formed on a surface of the wall, wherein the camera sensor is disposed in a predefined perspective position with respect to the wall, and wherein performing the position determination for the measuring device relative to the wall comprises: recording camera data of the camera sensor during the time the sensor data of the sensor unit is recorded by the measuring device, wherein the camera data depicts the measuring device positioned on the wall and the position marker disposed on the wall; determining a relative position of the measuring device relative to the position marker based on the camera data; and determining the position of the measuring device relative to the wall based on the relative position.

4. The method of claim 1, wherein performing the position determination further comprises: performing a time synchronization between recording the sensor data of the sensor unit and recording the camera data of the camera sensor; and determining the position information of the measuring device relative to the wall for each piece of sensor data of the sensor unit, taking into account the time synchronization.

5. The method of claim 1, wherein determining the position further comprises: determining a perspective of the camera sensor relative to the wall based on the predefined perspective position in which the camera sensor is disposed relative to the wall; and performing a perspective correction on the relative position of the measuring device determined based on the camera data relative to the position marker to account for the perspective position of the camera sensor relative to the wall and determining the position of the measuring device on the wall based on the perspective correction.

6. The method of claim 1, wherein the position marker is formed as an ArUco marker or a ChArUco marker.

7. The method of claim 1, wherein the position marker is formed by a marker formed on the wall and comprises: wallpaper patterns, light switches, outlets, windows, furniture.

8. The method of claim 1, wherein the camera sensor is disposed on a positioning device, and wherein the camera sensor is positioned via the positioning device in the predetermined perspective position relative to the wall.

9. A training dataset for training an artificial intelligence of a measuring device for wall diagnostics, wherein the training dataset was generated by the method for generating a training dataset for training an artificial intelligence for operating a measuring device according to claim 1.

10. A computer-implemented method for training an artificial intelligence of a wall diagnostic measuring device, comprising: providing a training dataset by performing the method for generating a training dataset according to claim 1; and training the artificial intelligence based on the training dataset to perform an object recognition of an object formed in a wall, wherein the object recognition comprises at least an object detection.

11. A computing unit configured to perform the method for generating a training dataset for training an artificial intelligence to operate a measuring device according to claim 1.

12. A computer program product comprising instructions which, when the program is executed by a data processing unit, prompt it to perform the method for generating a training dataset to train an artificial intelligence to operate a measuring device according to claim 1.

13. The method of claim 1, wherein the measuring device is a wall diagnostic device.

14. A computing unit configured to perform the method for training an artificial intelligence of a measuring device according to claim 10.

15. A computer program product comprising instructions which, when the program is executed by a data processing unit, prompt it to perform the method for training an artificial intelligence of a measuring device according to claim 10.
